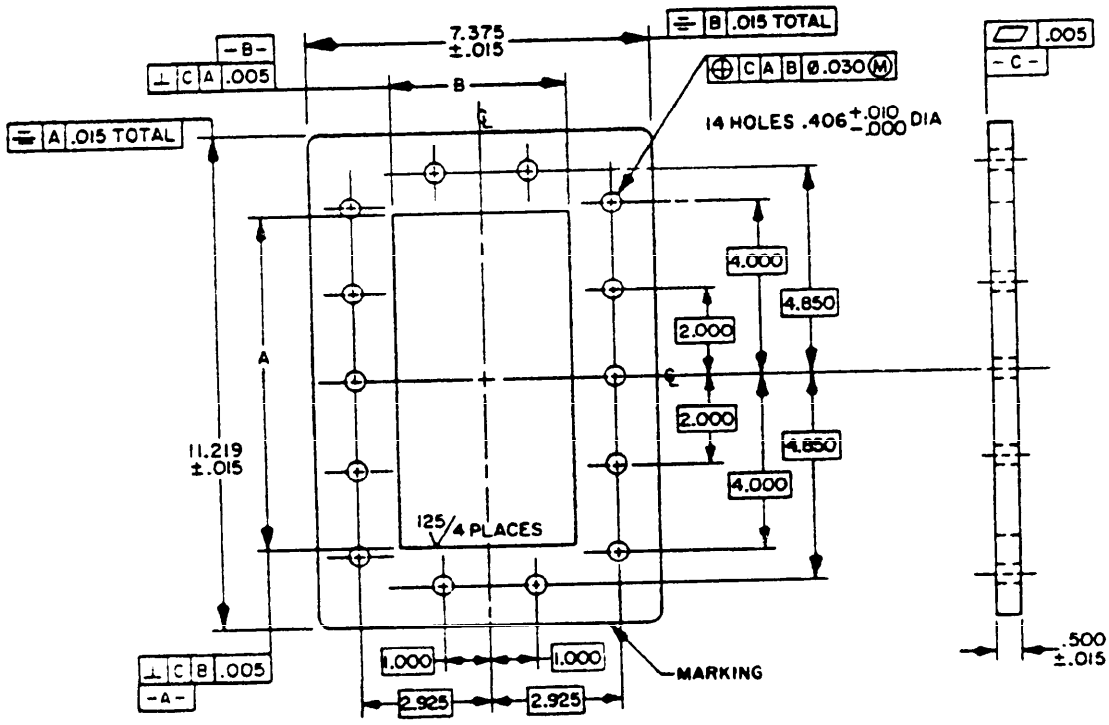
MILITARY SPECIFICATION

FLANGES, WAVEGUIDE, GENERAL PURPOSE, LARGE (UNPRESSURIZED CONTACT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide flanges described herein shall consist of this document and the latest issue of Specification MIL-F-3922.

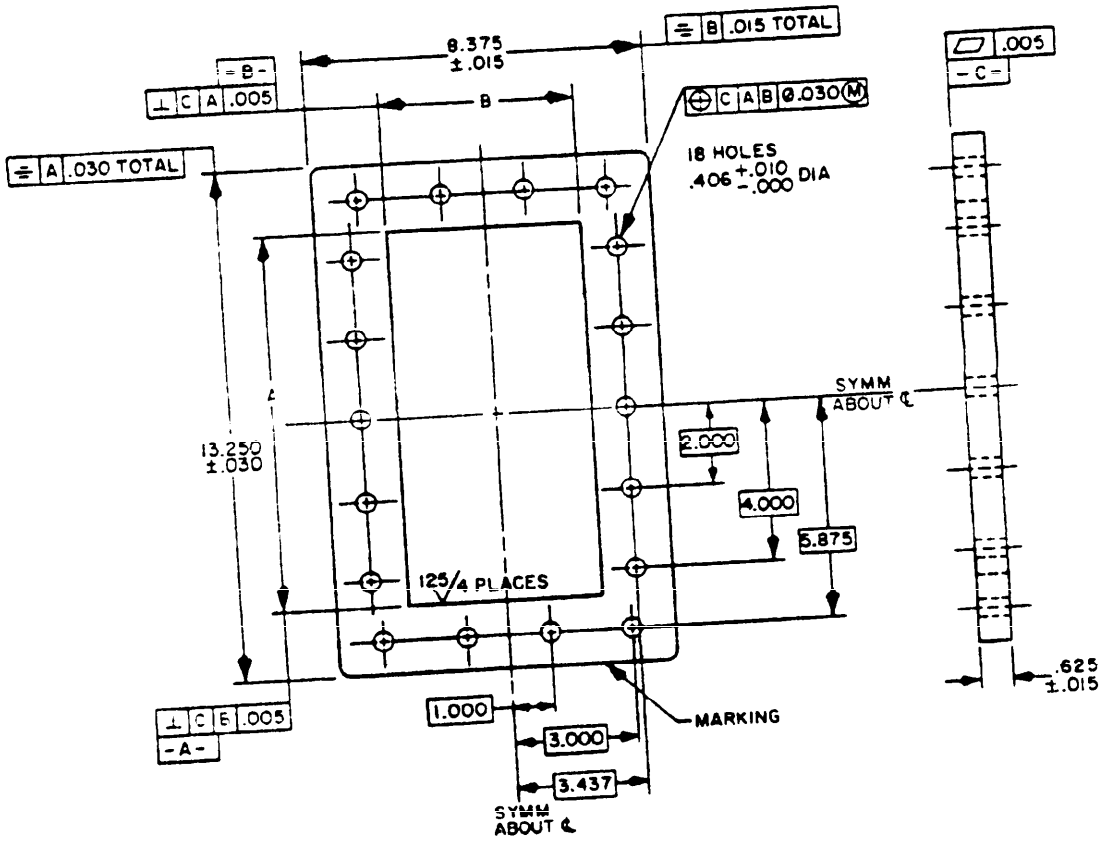


Configuration 1. Flange configuration for part number -01.

FIGURE 1. Flange.

FSC 5985

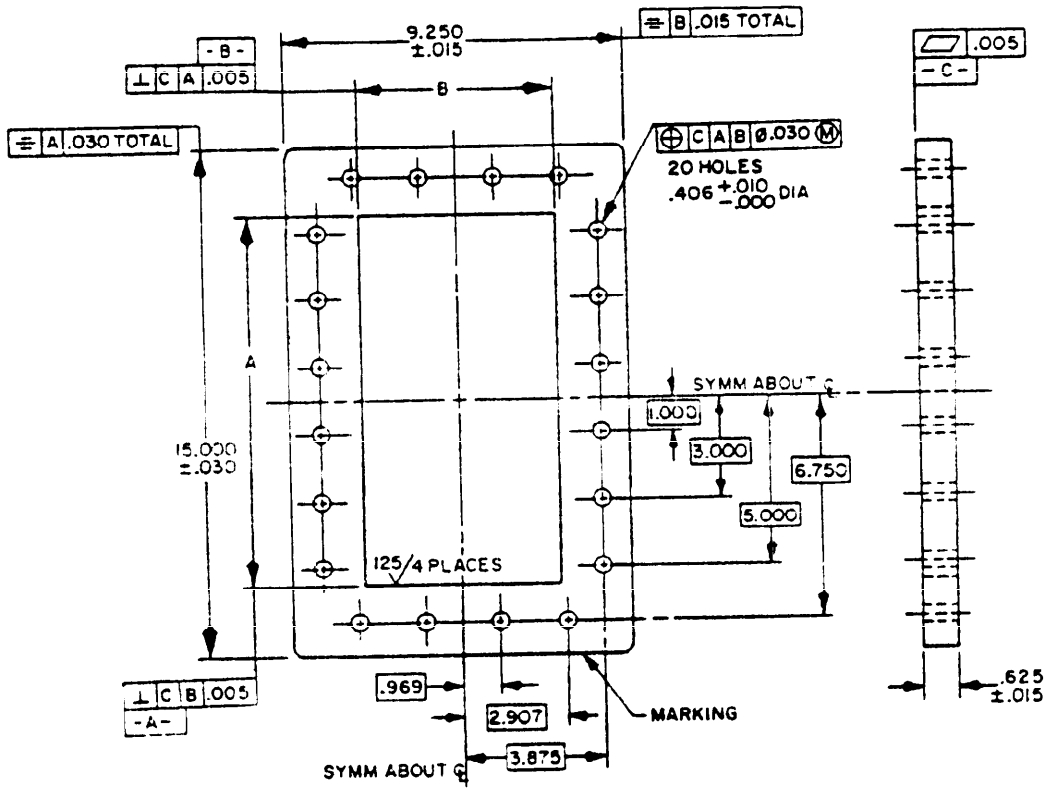
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Configuration 2. Flange configuration for part number -02.

FIGURE 1. Flange - Continued.

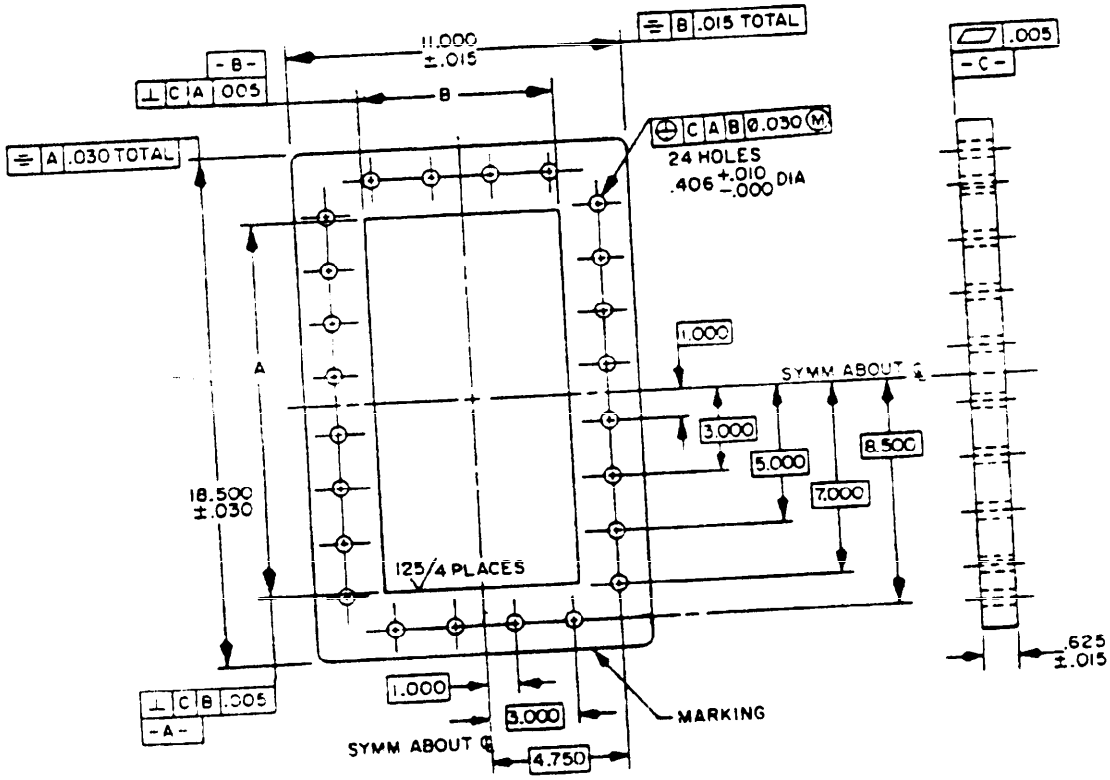
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Configuration 3. Flange configuration for part number -03.

FIGURE 1. Flange - Continued.

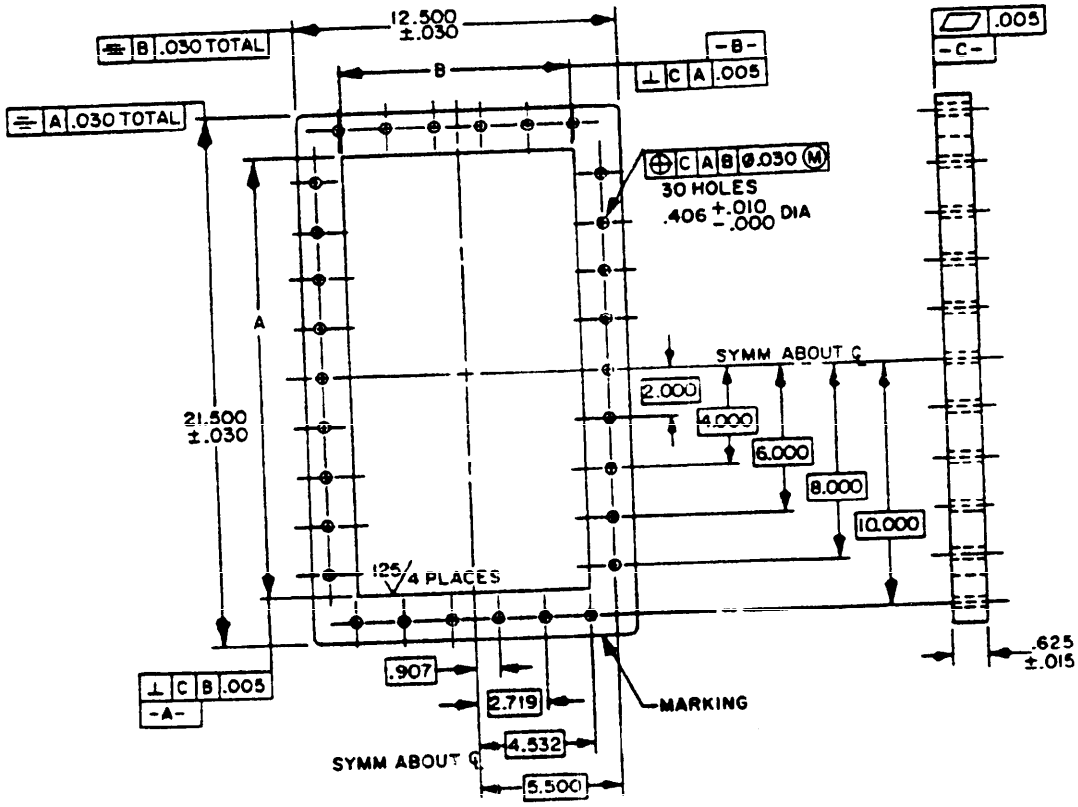
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Configuration 4. Flange configuration for part number -04.

FIGURE 1. Flange - Continued.

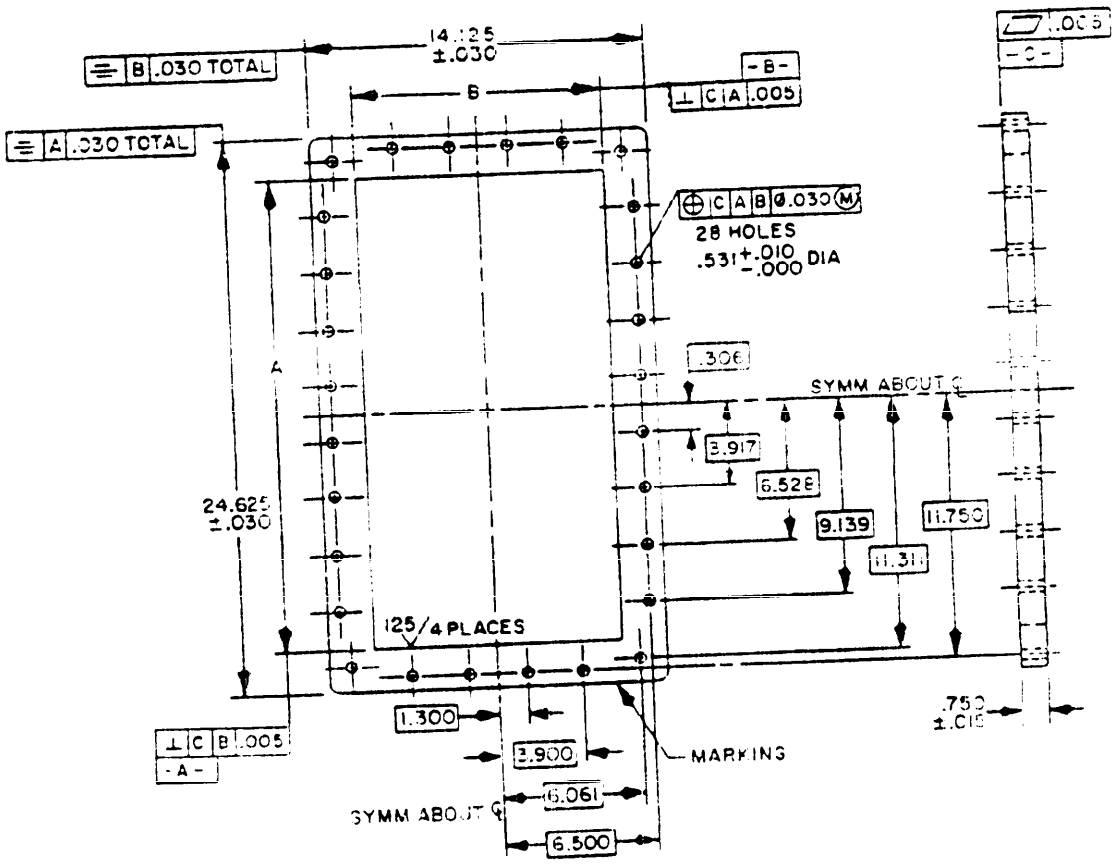
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Configuration 5. Flange configuration for part number -05.

FIGURE 1. Flange - Continued.

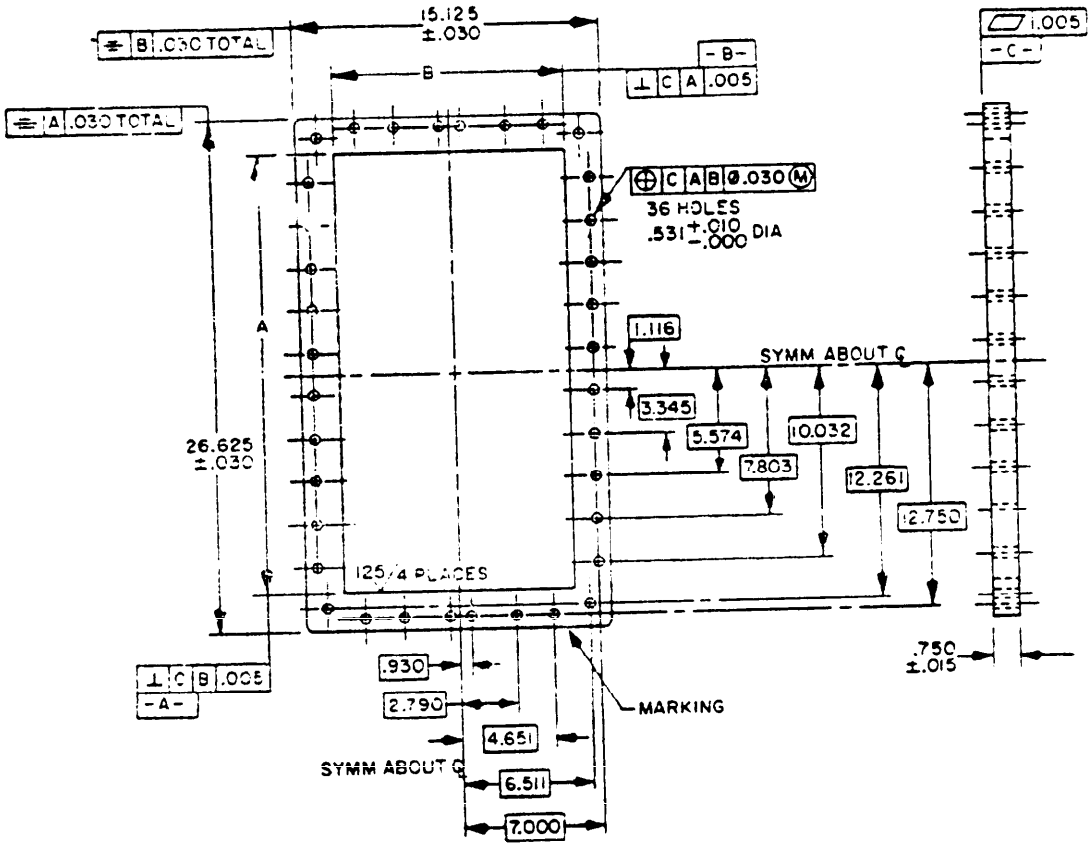
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Configuration 6. Flange configuration for part number -06

FIGURE 1. Flange - Continued.

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Configuration 7. Flange configuration for part number -07.

TABLE I. Part number characteristics and dimensions.

Part number M3922/76	Used with part no.		Dimensions	
	Waveguide MS3[illegible]-	Mating flange M3922/76	A +.005 (.13) -.000	B +.005 (.13) -.000
01	013, 014, 167	01	7.961 (202.21)	4.111 (104.42)
02	011, 012, 166	02	10.011 (254.28)	5.136 (130.45)
03	009, 010, 165	03	11.766 (298.86)	6.016 (152.81)
04	007, 008, 164	04	15.266 (387.76)	7.766 (197.26)
05	005, 006, 163	05	18.271 (464.08)	9.271 (235.48)
06	003, 004, 162	06	21.397 (543.48)	10.897 (276.78)
07	001, 002, 161	07	23.397 (594.28)	11.897 (302.18)

FIGURE 1. Flange - Continued.

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	1.300	33.02	4.532	115.11	6.750	171.45	11.311	287.30
.010	.25	1.306	33.17	4.651	118.14	7.000	177.80	11.750	298.45
.015	.38	2.000	50.80	4.750	120.65	7.375	187.32	12.261	311.43
.030	.76	2.719	69.06	4.850	123.19	7.803	198.20	12.500	317.50
.406	10.31	2.790	70.87	5.000	127.00	8.000	203.20	12.750	323.85
.500	12.70	2.907	73.84	5.500	139.70	8.375	212.72	13.250	336.55
.531	13.49	2.925	74.30	5.574	141.58	8.500	215.90	14.125	358.78
.625	15.88	3.000	76.20	5.875	149.22	9.139	232.13	15.000	381.00
.750	19.05	3.345	84.96	6.000	152.40	9.250	234.95	15.125	384.18
.907	23.04	3.437	87.30	6.061	153.95	10.000	254.00	18.500	469.90
.930	23.62	3.875	98.42	6.500	165.10	10.032	254.81	21.500	546.10
.969	24.61	3.900	99.06	6.511	165.38	11.000	279.40	24.625	625.48
1.000	25.40	3.917	99.49	6.528	165.81	11.219	284.96	26.625	676.28
1.116	28.35	4.000	101.60						

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Roughness of mating surfaces shall not exceed 63 microinches in accordance with ANSI B46.1, except that flaws shall be included in the roughness height measurement.

FIGURE 1. Flange - Continued.

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REQUIREMENTS:

Dimensions and configuration: See table I.

Material: Aluminum alloy.

Marking: See figure 1.

Items furnished with flanges: See table II.

TABLE II. Items supplied with flanges.

Part number M3922/76-	Hex head cap screw 1/	Hex nut	Flat washer 1/	Spring lockwasher 1/
01	.375-16 UNC-2A, 1.75 inch long, 14 each	.375-16 UNC-2B 14 each	.375 x .065 thick 28 each	.375 x .094 thick 14 each
02	.375-16 UNC-2A 2.0 inch long, 18 each	.375-16 UNC-2B 18 each	.375 x .065 thick 36 each	.375 x .094 thick 18 each
03	.375-16 UNC-2A, 2.0 inch long, 20 each	.375-16 UNC-2B 20 each	.375 x .065 thick 40 each	.375 x .094 thick 20 each
04	.375-16 UNC-2A, 2.0 inch long, 24 each	.375-16 UNC-2B 24 each	.375 x .065 thick 48 each	.375 x .094 thick 24 each
05	.375-16 UNC-2A, 2.0 inch long, 30 each	.375-16 UNC-2B 30 each	.375 x .065 thick 60 each	.375 x .094 thick 30 each
06	.500-13 UNC-2A 2.5 inch long, 28 each	.500-13 UNC-2B 28 each	.500 x .095 thick 56 each	.500 x .125 thick 28 each
07	.500-13 UNC-2A 2.5 inch long, 36 each	.500-13 UNC-2B 36 each	.500 x .095 thick 72 each	.500 x .125 thick 36 each

1/ Dimensions are in inches. Metric equivalents (to the nearest .01 mm) are given for general information only, and are based upon 1 inch = 25.4 mm. Metric equivalents are:

INCHES	MM	INCHES	MM
.065	1.65	.500	12.70
.094	2.39	1.75	44.45
.095	2.41	2.0	50.80
.125	3.18	2.5	63.50
.375	9.53		

Part number: M3922/76- (dash number from table I).

ENGINEERING INFORMATION:

Frequency range data: See table III. Gaskets for use with these flanges are covered by MIL-G-24211.

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TABLE III. Frequency range data.

Part number M3922/76-	Suggested frequency, range TE_{10} mode (GHz)
01	.960 - 1.45
02	.750 - 1.12
03	.640 - .960
04	.490 - .750
05	.410 - .625
06	.350 - .530
07	.320 - .490

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI,
Navy - EL, CS
Air Force - 11, 99
DSA - ES

User activities:
Army - MU
Navy - AS, MC, SH
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0836)

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